

A Novel Solvent-Based Recycling Technology: From Theory to Pilot Plant

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Abstract

We have developed a solvent-based technology for the extraction of single pure resins of food quality. The technology is called Solvent Targeted Recovery and Precipitation (STRAP). We have carried out numerous lab-scale experiments (in batches of 10-1000 g), with various plastic wastes, both post-industrial recycling and post-consumer recycling, including multi-layer flexible films. The key to the successful implementation of STRAP is the ability to pre-select solvents and temperatures to selectively dissolve a single polymer from all components in the mixture. The team has developed an extensive fundamental understanding of dissolution, enabling a successful operation. To aid in solvent selection, we developed a first-principles molecular modeling approach to rapidly predict temperature-dependent polymer solubilities. The technology is currently being scaled up to a 0.5 ton/day continuous Process Development Unit (PDU), which will be the basis for the next industrial scale. The technology requires relatively low temperatures (100-135°C) and equipment that is available in the chemical industry. We carried out extensive techno-economic and life-cycle assessments that showed (i) a STRAP system can be profitable as a stand-alone system at a 4,000 ton/y facility, with a great advantage of size, and (ii) due to the low process energy, the GHG emissions are reduced by over 90% in comparison to fossil-based polymer production. A line of the STRAP technology extracts a single polymer; to extract multiple polymers, multiple STRAP lines are required. Various methods can remove ink, pigments, or colorants, producing pure, colorless resins of food quality. The STRAP scheme depends on the source of the waste. It is to be noted that we successfully separated nine resins from arbitrarily mixed plastic wastes. Through our industrial partners, we showed that our extracted resins can be reused for their original film applications. The single most important economic factor is solvent recovery, at >99.9% of the solvent used, which is the solvent recovery system we are developing. Both lab scale and upscale results are presented and discussed.

Keywords: Solvent-based recycling, molecular modeling, scale up, multi-layer plastics, techno-economic analysis, life-cycle analysis

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37.1 Introduction

In the US, most waste plastics are either landfilled or incinerated causing negative environmental impacts [1]. Recently, substantial research activity has been devoted to developing techniques to identify, separate, and reuse components of plastic waste [2]. However, the complex composition of mixed plastic waste poses challenges to existing separation techniques.

We have developed a novel *solvent-based* approach for the preferential dissolution of polymer resins in respective solvents that extract the resins in pure form, retaining their original properties [3, 4]. The recovered resins can thus be reused in their original applications. The process extracts only the respective polymer resins removing additives, ink, pigments, and adhesives. This process is referred to as the Solvent-Targeted Recovery and Precipitation (**STRAP**) process, with a pending patent application.

Some attempts have been made to commercialize solvent-based plastic recycling [5]; The main ones are (1) *Multi-cycle* [6] by Fraunhofer Inst., (2) *PureCycle* [7], and (3) *Newcycling*[®] by APK AG [8]. MutliCycle[®] has been licensed to Unilever, Polystyrene loop, and Circular Packaging. These efforts have faced challenges: Unilever built a 3-ton/day pilot plant in Indonesia but due to operational issues stopped the plans for scaling up the technology [9]. APK AG built an 8 kton/yr solvent-based recycling pilot plant to produce PP and PE from post-industrial waste multilayer plastics [10]. APK's technology is based on dissolving a plastic using a solvent mixture from a group of alkanes, isooctane, or cycloalkanes [11]. No reports have indicated why these technologies struggled to become commercial, but this is common with pioneer process technologies.

The key to the successful implementation of our STRAP is the ability to pre-select solvents and temperatures capable of selectively dissolving a single polymer from all components in the plastic mix. The team has developed an extensive and expansive fundamental understanding of dissolution, enabling a successful operation. To aid in solvent selection, we developed a *first-principles molecular modeling approach* to rapidly predict temperature-dependent polymer solubilities in solvents [12]. Our approach is based on a careful selection of the resin-solvent system and refined process parameters, temperature, concentration, residence time, and careful filtration of inks and colorants. STRAP-produced resins are pure and white, with no ink or colorants. We have made many lab-scale successful attempts to separate polymers from various wastes, such as Flexible multi-layer films, mixed plastic wastes, post-industrial recycling, post-consumer recycling, MRF residues, and municipal solid wastes. Our intention is to develop the technology that can reach the market in a relatively short time.

In this paper, we will present results from (1) our molecular modeling work for the selection of the specific solvent for the specific resin; (2) lab-scale results, specifically for multi-layer films provided by Amcor; and (3) our scaling-up efforts.

37.2 Main Content of Chapter

37.2.1 Molecular Simulations and Experimental Verification

The team has developed an extensive fundamental understanding of dissolution that enables a successful operation. We developed a first-principles molecular modeling approach to

rapidly predict temperature-dependent polymer solubilities in solvents [3, 12]. A series of computational methods are used to evaluate polymer solubilities and compare them to experiments for validation. Figure 37.1 shows the typical steps – Step 1: A molecular dynamics (MD) simulation of a single oligomer is performed in a dilute solution to obtain a simulation trajectory of various oligomer conformations; Step 2: A set of conformers is selected from the MD trajectory to span a range of representative oligomer structures; Step 3: Density functional theory (DFT) calculations are performed for the selected conformers to obtain corresponding screening charge density profiles; Step 4: Screening charge density profiles are input to Conductor-like Screening Model for Real Solvent (COSMO-RS) solubility calculations; and Step 5: Experimental solubility measurements are used to benchmark the computational predictions. We provide details on each step in the sections below.

In the first step, we model target polymers as oligomer molecules and perform MD simulations of these oligomers in dilute solutions (with both good and poor solvents) to obtain trajectories of representative oligomer conformations [12]. Solvents were selected based on literature reports of good and poor solvents for the target polymers. We then sampled representative oligomer structures (referred to as conformers) from the MD trajectories based on two structural parameters: the radius of gyration (R_g) and the solvent-accessible surface area (SASA). Sampling conformers that cover a range of values of these two parameters can provide reliable input for COSMO-RS solubility estimations [12, 13]. Based on extensive analysis of tradeoffs between prediction accuracy, oligomer size, and a number of conformers, we adapted a workflow in which 20–30 conformers were sampled for each polymer for oligomers comprised of ~6 repeat units. The selected conformers were input to DFT calculations to obtain screening charge densities (COSMO files). The DFT calculations included a geometry optimization in implicit water using the conductor-like polarizable continuum model and a single-point calculation in the infinite dielectric constant limit.

COSMO files from the DFT calculations were input to COSMO-RS for solubility calculations. COSMO-RS predicts the thermodynamic properties of multicomponent systems based on quantum mechanical calculations and statistical thermodynamics methods [14, 15]. The chemical properties of each molecule are represented by the probability distribution of the screening charge densities (called the σ -profile). σ -profiles of all oligomer conformations with deactivated terminal groups were used to approximate the σ -profile of the corresponding polymer [16]. The σ -profiles were then used to calculate the chemical potential of the polymer to enable predictions of solubility via a solid-liquid equilibrium calculation. This calculation requires the polymer melting temperature and an experimentally measured solubility as reference input. Experimental reference inputs were measured

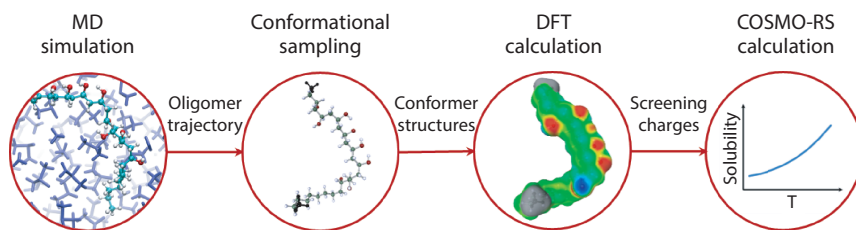


Figure 37.1 Flow diagram summarizing the methods for computational polymer solubility prediction and experimental validation.

for each target resin and corresponding melting temperatures were taken from literature sources. With this approach, we were able to obtain predictions of temperature-dependent solubilities to compare against experimental values.

We performed multiple tests to assess the accuracy of computational predictions. Table 37.1 compares predicted and experimentally measured solubilities for a PE sample in various common solvents, with temperatures listed in the table [12]. We utilized the experimental solubility of PE in toluene at 110°C as a reference value for the solid-liquid equilibrium calculation for all solvents. As expected, our computational protocol predicts near-zero solubilities for all solvents other than toluene in good agreement with the experimental measurements.

Notably, several of these solvents are predicted by some sets of Hansen Solubility Parameters to be good solvents (e.g., hexane, THF, and ethyl acetate), indicating that our computational approach qualitatively outperforms conventional methods for solvent selection based on Hansen Solubility Parameters in addition to providing quantitative solubility estimates.

We next assessed our ability to predict the solubility of EVOH (a common component of multilayer plastic films), which is a challenging task because EVOH is a copolymer that has two different monomer units. The arrangement of these units leads to different types of structures, so we assessed prediction accuracy using three different structural arrangements of the EVOH repeat units Table 37.2 compares solubility predictions to the experimental data for each of these 3 structures. For this comparison, the experimental solubility of EVOH in DMSO at 95 °C was used as the reference for the solid-liquid equilibrium calculation. Clearly, DMF was recognized as a good solvent in all three cases, but only the block structure result correctly identifies it as the best solvent in the list. Solubilities are somewhat over-predicted for experimentally identified nonsolvents, with ethanol being the most notable deviation. However, the values clearly differentiate solvents and nonsolvents, with a slightly higher threshold (3 wt%) required to distinguish these categories than is necessary for PE. With respect to quantitative predictions, the random structure has the lowest overall mean prediction error, which is consistent with the fact that EVOH is a random copolymer [17].

Table 37.1 COSMO-RS solubility predictions for PE compared to experimental measurements. Toluene is used as a reference solvent and hence its prediction is exact.

Solubility (wt%)			
Solvent	Temp (°C)	Experiment	Simulation
DMSO	95	0.040	0.34
Water	95	0.002	0.00
THF	65	0.018	0.54
Hexane	65	0.031	0.85
Ethyl acetate	77	0.004	0.52
Ethanol	78	0.011	0.20
Acetone	55	0.001	0.06
Toluene	110	14.56	14.56
DMF	100	0.009	1.85

Table 37.2 COSMO-RS solubility predictions for EVOH compared to experimental measurements. Three different copolymer structures are compared (Figure 37.2).

Solvent	Temp (°C)	Experimental solubility (wt%)	COSMO-RS solubility predictions		
			Block (wt%)	Alternating (wt%)	Random (wt%)
DMSO	95	24.02 ^[a]	24.02	24.02	24.02
Water	95	0.00	0.01	0.03	0.03
THF	65	0.02	5.53	2.06	1.78
Hexane	65	0.00	0.02	0.00	0.00
Ethyl acetate	77	0.05	2.47	0.66	0.49
Ethanol	78	0.00	7.63	3.94	2.98
Acetone	55	0.00	1.81	0.58	0.45
Toluene	110	0.00	4.49	0.64	0.39
DMF	100	30.63	27.49	21.19	20.48

^[a] experimental data in this solvent was used as reference for the solid-liquid equilibrium calculation, so all three structures have the same calculated solubility.

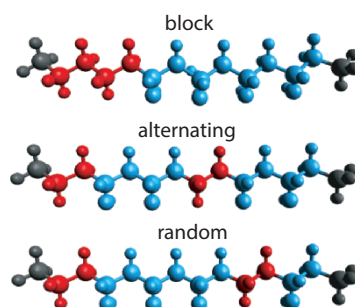


Figure 37.2 Three types of structures of EVOH oligomer with 2 ethylene units (red) and 4 vinyl alcohol units (blue). Oligomers are modeled to have end methyl groups (black) that are deactivated such that they do not contribute to the COSMO-RS calculations.

We also utilized EVOH to assess the capability of our protocol to assess temperature-dependent variations in solubility. Figure 37.3 shows the predicted EVOH solubility in DMSO as a function of temperature. Experimental data at four temperatures are included for reference. Predictions were based on either 1, 5, or 20 EVOH conformers, with 10 separate subsets of conformers used for predictions. Predicted solubilities with only a single conformer varied substantially and lie with the region indicated by the light blue color. The region bounding the solubility predictions decreased substantially for the 5-conformer sets (light green region). For the 20-conformer sets, the solubility calculations are highly consistent leading to a deep blue region that nearly converges to a solid line such that variations are difficult to discern by eye. These results indicate that increasing the number of conformers both leads to a decrease in prediction accuracy and decreases the variation between different subsets of conformers, suggesting more robust results.

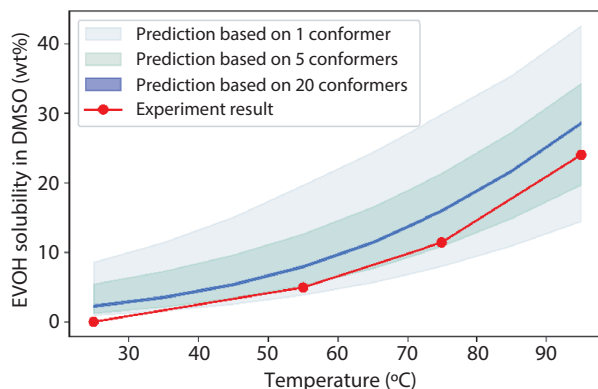


Figure 37.3 Temperature-dependent solubility prediction for EVOH in DMSO with different conformers.

Moreover, all conformer sets exhibit reasonable agreement with experiments, with the final results (for 20 conformers) slightly overestimating the measured solubilities but exhibiting similar trends.

Based on the good agreement with experimental data, we expanded the predictions to assemble a database of polymer solubilities. In total, we predicted the solubilities of 8 common polymers (PE, PP, PS, PET, PVC, Nylon-6, Nylon-66) in 1007 solvents. For each solvent, the solubility of each polymer was computed at both room temperature and a higher temperature to suggest bounds on feasible solubilities. This database permits rapid identification of solvents for STRAP based on the criteria that a solvent must selectively dissolve exactly one polymer from a multicomponent system; specifically, the solubility should be high for a target polymer but low for all other polymers.

Table 37.3 shows examples of screening for the separation of PE and EVOH by presenting 9 selective solvents and corresponding temperatures. Further examples of the application to real STRAP processes are described in the sections below. Overall, a strong computational tool for solubility predictions has been developed for polymer-solvent combinations and it has successfully demonstrated good agreement with the experiment. The applicability of the approach was demonstrated by identifying selective solvents for various polymers from a library of 1007 solvents.

37.2.2 Lab-Scale STRAP Results for Multi-Layer Flexible Films

STRAP was applied to a printed flexible multilayer plastic film to recover and produce clear polymer resins after incorporating a deinking step. The target film is a post-industrial oriented polyester film (OPET) that comprises PE, EVOH, and PET with polyurethane (PU) inks (Figure 37.4). A challenge with this specific plastic waste feedstock is how and at what stage of the process should the ink components be removed. Different surfactant solutions and PU-selective solvents were considered to achieve the recovery of the polymer fractions without any coloration.

The flexible printed multilayer film composed of PE, EVOH, PET, and PU-based inks was processed through STRAP to recover all polymer components. Figure 37.5a shows solvents, temperatures, and dissolution times for each step. In our approach, the PE and

Table 37.3 Computational solvent screening for PE and EVOH. Nine selective solvents and corresponding temperatures are presented as examples (no PE-selective solvents were identified at room temperature).

Solvent	Temp (°C)	PE solubility (wt%)	EVOH solubility (wt%)	Comment
Hydrazine	25	0.00	26.38	Room temperature, EVOH-selective
Ethylenediamine	25	0.00	24.40	
n-butylamine	25	0.00	15.26	
Octane	124.6	45.40	0.13	Near boiling point, PE-selective
1-octene	120.2	35.22	0.21	
Tetrachloroethylene	120.3	21.34	0.08	
Methylhydrazine	86.5	0.23	34.21	Near boiling point, EVOH-selective
Acetic acid	116.9	0.62	25.46	
Formic acid	100	0.02	24.42	

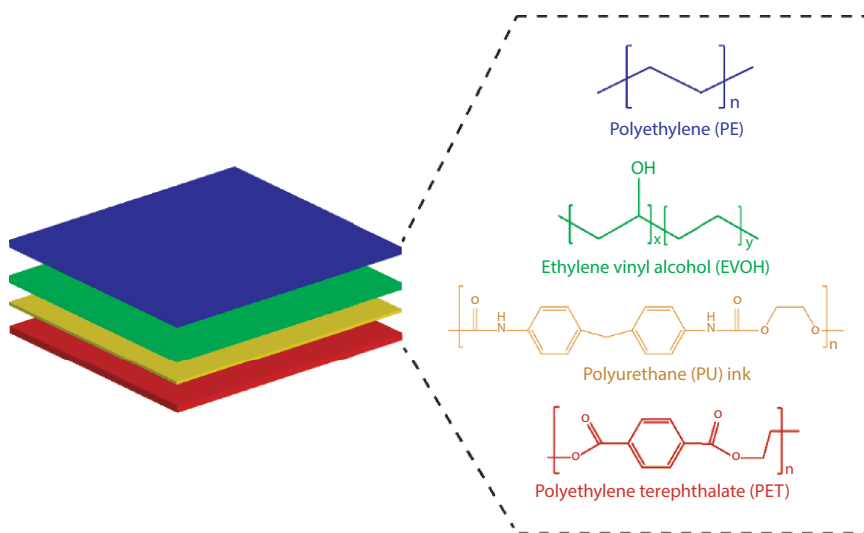


Figure 37.4 OPET multilayer printed film composed of PE, EVOH, PET, and PU-based inks, manufactured by Amcor.

EVOH are dissolved and recovered, leaving the PET and ink components behind. After this, the ink is removed from the PET component using GVL. Polymer precipitation was achieved by reducing the temperature of the respective solvent and not by the addition of antisolvents. In our previous work, this was demonstrated to be beneficial both economically and environmentally. On visual inspection, the final recovered polymers after

STRAP exhibited little to no coloration (Figure 37.5b). The ink removal was carried using gamma-Valerolactone (GVL) to separate white, black, and yellow ink from PET of a printed multilayer film (Figure 37.5c).

Table 37.4 shows the yield of each component, including the ink residue (black and white) recovered after the ink removal step. Due to the high percentage of PE in the printed film, an additional PE dissolution step was required to recover most of the polymer and clean out the experimental setup. This additional PE dissolution step was done in every experiment in Table 37.4, which allowed for an average PE yield of 61.8 wt%.

We found that the material that was not recovered after STRAP was left behind in our equipment, mostly in the round bottom flasks and filters. The overall mass balance improved when accounting for the material lost in the equipment which was around 8 wt% of the plastic feed. Furthermore, part of the ink (mostly yellow color) can be recovered via distillation of the GVL, around 3 wt%. The overall mass balance can be >95 wt% of the starting material, considering the average recovery of PE, EVOH, and PET. This gives indication that when dealing with flexible plastics that have been shredded, material will

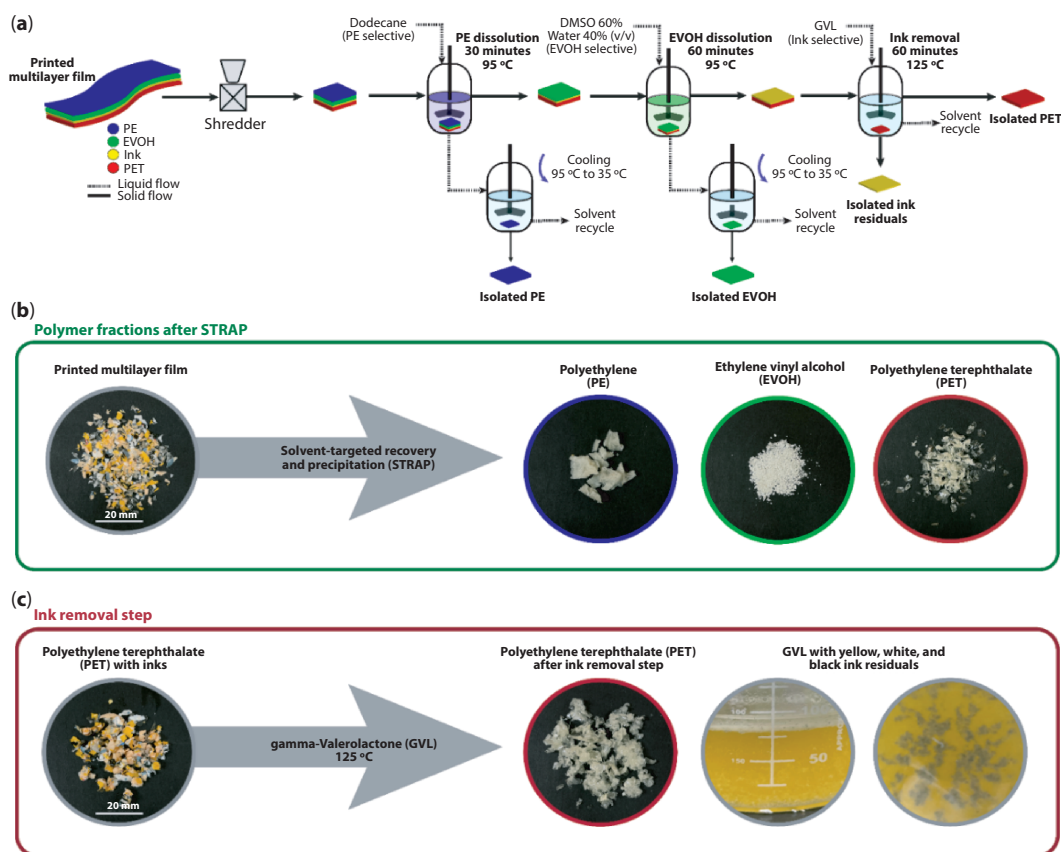


Figure 37.5 (a) Solvent-targeted recovery and precipitation (STRAP) of a printed multilayer film composed of PE, EVOH, PET, and PU-based inks, (b) Photos of each polymer after the STRAP, (c) STRAP deinking step with gamma-Valerolactone (GVL) to separate white, black, and yellow ink from PET of a printed multilayer film.

Table 37.4 Resin yield: printed multilayer film.

Exp	Component yield (wt%)			
	PE	EVOH	PET	Ink
1	59.8	7.70	19.8	1.4
2	62.4	6.94	17.6	2.1
3	63.2	7.74	18.2	0.6
Avg.	61.8	7.46	18.5	1.4
SD (σ)	1.7	0.45	1.2	0.8

be left behind in the equipment which should be considered when thinking about larger scale systems. Furthermore, it was challenging for us to predict and measure consistent amounts of material lost in the equipment since it depends on multiple factors like plastic size, stirring rates, and it is subject to how the material is handled by the person conducting the batch experiment.

Previous STRAP experiments with transparent multilayer films were initially demonstrated with film sizes of 1x1 cm and this necessitated extensive dissolution times to recover the target polymers. We investigated the effects of film sizes over the dissolution time of the PE component in the printed multilayer film. This was tested in the PE-dissolution step using dodecane at 95°C and 500 RPM. Initial steps with a dissolution time of 30 minutes yielded 61.78 ± 1.74 wt% of PE when the film sizes were > 3 mm. This yield was calculated as the amount of extracted PE over plastic feed in a single experiment. Additional size reduction of the multilayer films to 3 mm and 1 mm showed improvement in the PE dissolution time. At a total PE dissolution time of 8 minutes, the PE recovery improves by reducing the size of the films to 3 mm. From 3 mm to 1 mm, a slight improvement in the PE yield was observed. The 8 minutes of dissolution time were divided in two steps, a 5-minute step and a 3-minute step. As explained in the experimental demonstration section, multiple steps were carried out to recover most of the dissolved PE due to its high composition in the film, and this helped with cleaning out the experimental setup. From the above results, we concluded that by downsizing the plastic material to the proper size (1-3 mm) and having vigorous mixing and the required dissolution temperature, we were able to extract most of the PE with dodecane. In dissolution-based plastic recycling technologies, plastic sizes become important whenever short dissolution times are of interest.

37.2.3 Polymer Characterization

We use IR spectra for resin identification. In Figure 37.6a, the IR spectra of the PE STRAP and PE virgin resin are compared, displaying identical spectral bands with no trace of any other bands belonging to other resins, indicative to the purity of the extracted PE. Figure 37.6b and Figure 37.6c shows similar characteristics for EVOH and PET, respectively. The molecular weight values of the recovered PE from the printed multilayer film by STRAP were determined with high-temperature gel permeation chromatography (HT-GPC).

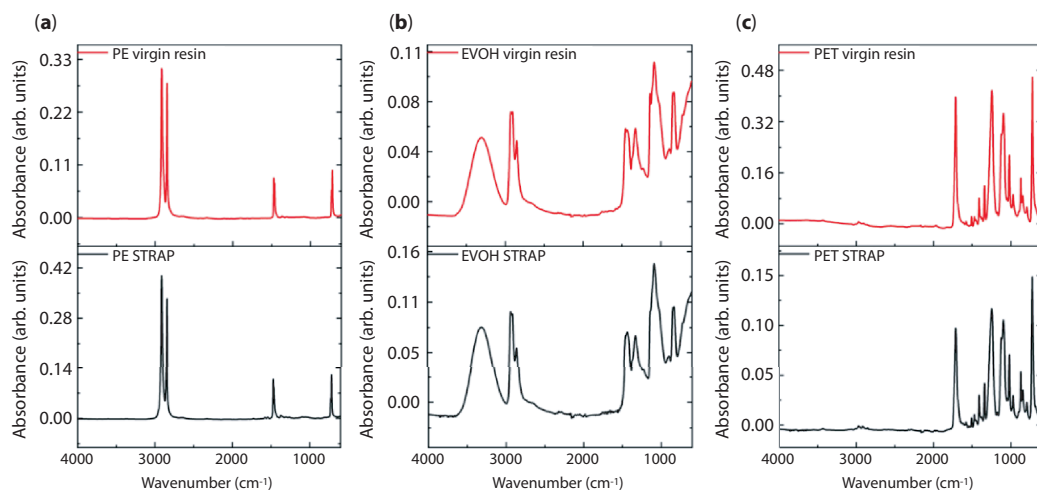


Figure 37.6 ATR-FTIR spectra of virgin resins and polymers recovered by STRAP from the printed multilayer: (a) PE, (b) EVOH, and (c) PET.

The printed film itself was manufactured with both LDPE and LLDPE, and the produced PE from STRAP had molecular weight values that were within the range of the pure resins. This gives evidence that the molecular weight of the PE is not significantly affected after being treated with dodecane in the STRAP process. The recovered PE, EVOH, and PET for the most part had similar thermal properties to the corresponding pure virgin resins used in the starting material.

For example, as seen in Table 37.5, the melt temperature ($T_{m,2}$) of the recovered polymers from the printed film by STRAP were comparable to the virgin resins. Similar results were obtained by Cecon *et al.* [18] for the PE resins recovered using STRAP, with the thermal properties within the ranges displayed by the virgin HDPE and LLDPE resins used in a rigid transparent multilayer film. However, differences were observed in some parameters of the PET STRAP samples, possibly due to residual polymers or ink components. This

Table 37.5 Thermal and molecular parameters for virgin resins and polymers recovered from the printed multilayer film by STRAP.

Resin	T _c (°C)	T _{m,2} (°C)	ΔH _c (J/g)	ΔH _{m,2} (J/g)	Crystallinity
PE STRAP	105.0	119.7	76.0	82.3	28.38%
LDPE Virgin	98.3	112.0	84.8	86.1	29.69%
LLDPE Virgin 45G	106.1	122.4	83.3	85.5	29.48%
LLDPE Virgin 47N	107.0	122.2	70.2	73.3	25.28%
EVOH STRAP	150.4	175.7	41.8	37.4	17.17%
EVOH Virgin	147.8	176.4	54.2	54.3	24.93%
PET STRAP	209.6	246.0	41.7	22.7	16.21%
PET Virgin	169.2	244.6	30.0	38.5	27.50%

could be addressed by carrying out additional ink removal steps to ensure PET is free of any contaminants. After STRAP, decreases in the crystallinities were observed for the recovered EVOH and PET, which could also be an indication of contamination in the samples. Our previous characterization study of STRAP recovered materials did not determine significant changes in the crystallinity but in the melt temperature of PET, which was associated with solvent retention in the polymer matrix as PET was the main layer in the film.

37.2.4 Scaling-Up

Figure 37.7 is a 3-D depiction of one scaled-up STRAP line at 25 kg/hr, i.e., one solvent and one resin extracted. The shredder (1) downsizes the plastic waste into 2-4-mm flakes with an aspect ratio of 1. The screw conveyor (2) transfers the material to a live bottom hopper (3). The bucket elevator (4) conveys the material into a dissolution tank (5), with an actuated control valve. The tank is heated to the desired temperature (e.g., LDPE at 110°C), and the solvent is fed from a tank (10).

This tank contains a nitrogen blanket for safety. After 5-6 min for complete dissolution, the blend is drained by opening the ball valve at the bottom of the tank and fed into a hot filter (6) which separates the solution from the non-dissolved flakes. The temperature of the hot filter is controlled by a heater. The non-dissolved flakes flow through a conveyor (13) and a control valve into storage tank. The plastic solution flows from the dissolution tank into a precipitator (8) that separates the pure resin from the solvent by cooling, using a chiller (11). The mix from the precipitator flows into a cold filter (12) where the solvent flows into a solvent tank (10) and the resin is stored in storage tank. We will describe the critical components of this system.

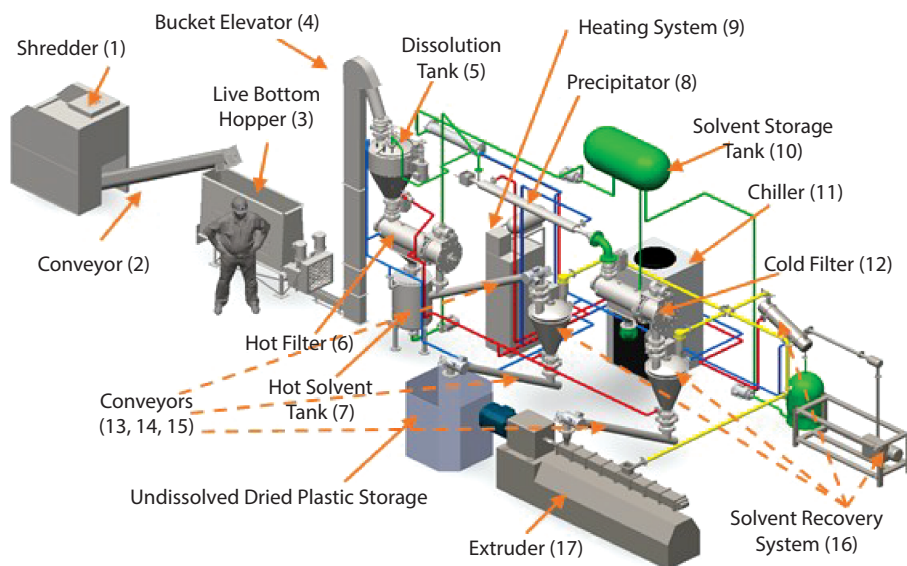


Figure 37.7 3-D presentation of the STRAP pilot unit.

37.2.5 Plastic Waste Dosing

We developed an accurate feed bin at a rate of up to 4 kg/min (Figure 37.8) with negative angled walls. The bin was tested with wastes for thousands of hours, including pre-processed and shredded plastics, encountering no bridging or flowability issues.

37.2.6 Dissolution Tank

Figure 37.9 shows the dissolution tank where the plastic flakes are flown into the tank through actuated ball valves, and the tank is filled with hot solvent. A shaft rotates at ~ 1700 rpm with two 5-in impellers. Baffles were added at the tank walls to create turbulent flow. Figure 37.10 shows the top view of the plastic-solvent flow in the tank, with/without baffles. The flow without baffles is laminar and with the baffles is turbulent. The solvent-to-plastic ratio is 7:1. The dissolution time has been extensively studied, as a function of impeller type, rotation frequency, and temperature. Figure 37.11 shows results for PE dissolution yield (calculated as dissolved plastic solvent over the initial plastic amount). As seen, complete dissolution requires about 20 seconds, which is indicative of excellent mixing with no mass transfer limitation.

37.2.7 Hot Filter

The hot filter separates the solution from the non-dissolved residues. We purchased a commercial filter that is based on centrifugal separation, shown in Figure 37.12. The undissolved plastics and the solution mix flows from the tank through a 4-in actuated valve into the filter. In the first section, a short screw auger conveys the mix into the centrifugal section with a still screen, with 4 axially angled brushes rotating at 800–1000 rpm. The solution is centrifugally “pushed” through the screen in a radially outward direction, whereas the non-dissolved plastic flakes are conveyed axially by four paddles towards the outlet. Note that the filter is heated using a jacket to avoid early precipitation.

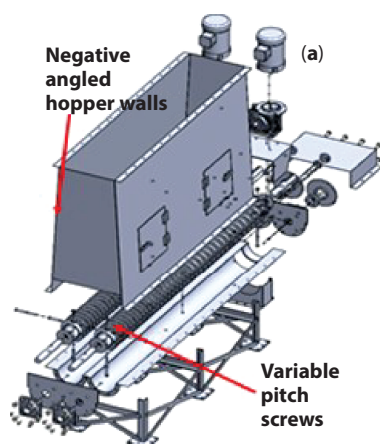


Figure 37.8 The hopper.

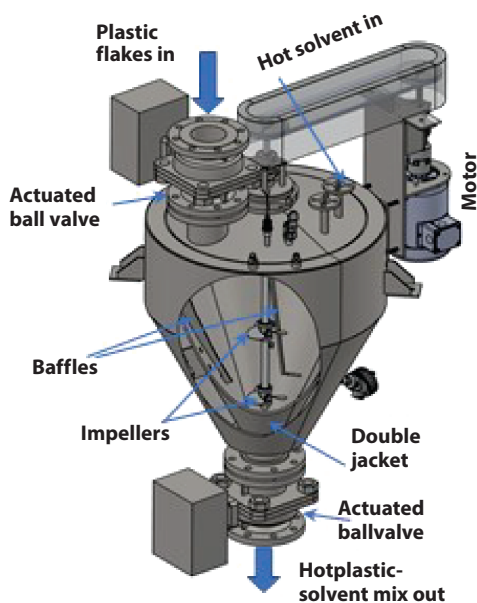


Figure 37.9 The dissolution tank.

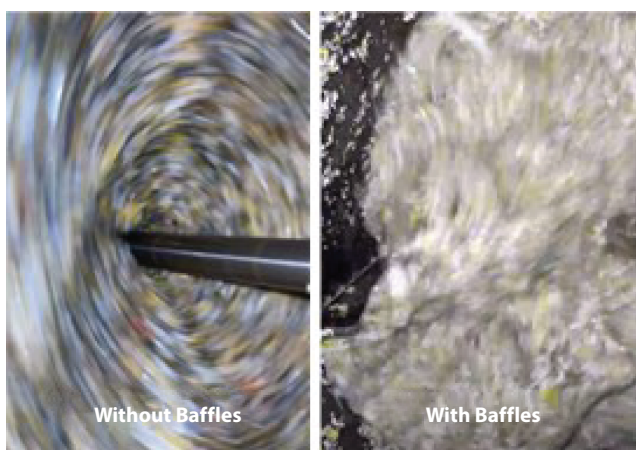


Figure 37.10 Flow patterns with/-out baffles.

We carried out numerous runs, measuring: (i) the solution outlet weight, (ii) the non-dissolved plastic weight, (iii) filtration efficiency, and (iv) filtration dynamics. We carried out numerous drainage experiments, where the 4-in ball valve of the dissolution tank was opened (~ 1 s). The plastic-solvent mix was drained through the valve directly into the hot filter. We measured the outlet from both the solid and liquid ports (Figure 37.13). Figure 37.13 shows typical results of drainage dynamics, as solid outlet vs. time. A drainage rate equivalent of ~ 5 kg/min was measured, which is a fast drainage rate. The filter can operate for extended periods without maintenance due to the active brush mechanism continuously wiping the filter screen. The status of our filtration technology is that the solid

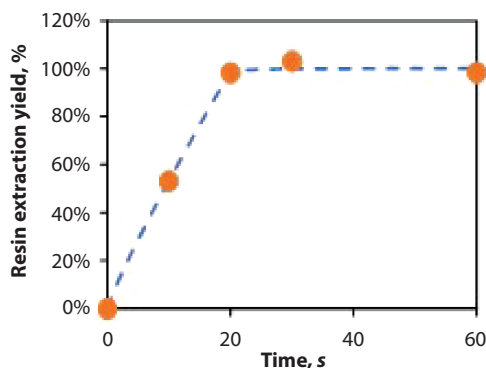


Figure 37.11 Dissolved resin vs. time in dissolution tank.

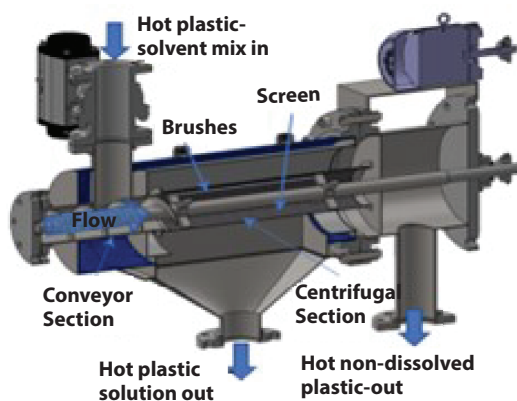


Figure 37.12 The filter.

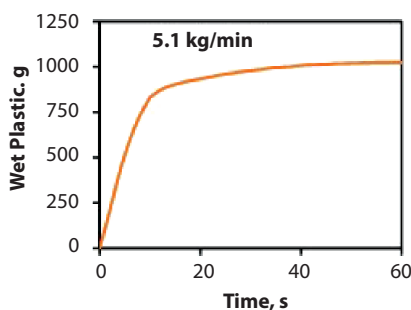


Figure 37.13 Hot filter plastic outlet.

material after filtration comprises 10% solvent, noting that this is equivalent to recovering **99% of the solvent**.

37.2.8 Precipitator

The precipitator receives hot plastic solution from Filter 1, cools it to precipitate the plastic. We note that key challenges are required temperature for full precipitation, plastic tendency

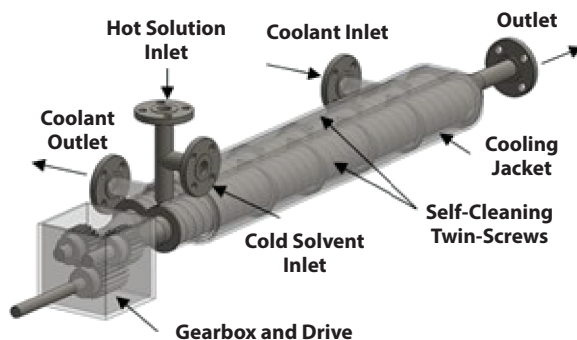


Figure 37.14 The twin-screw precipitator.

to stick to surfaces, and the time required to reach precipitation temperature. We used our own solubility data to design the temperature required for precipitation [3, 12]. Molecular simulations predicted the solubility (validated by lab experiments [3, 12]) From our experience, twin-screws, shown in Figure 37.14 have a self-cleaning mechanism as the screw flights scrub each other and housing walls, generating clean surfaces. The solution in the twin-screw, flows from one screw to the other, generating excellent mixing, with fast heat transfer, such that a 1.5-m long, 2-in diameter, can cool down the solution to $\sim 35^{\circ}\text{C}$ at the outlet in <10 min.

37.2.9 Solvent Recovery and Solvent Purification

We are currently building a solvent recovery device that can recover the solvent at a rate of 99.9%. As for purification, the solvents may have residual resins, inks and colorants and require purification. Solvents are used and recovered extensively in chemical industry without any indication for degradation. Degradation occurs only if a chemical reaction occurs with the solvent. In STRAP, there are no reactions involved, unless some chemicals in the plastic wastes react with the solvent. Part of the current program is to monitor the quality of the solvent by IR spectroscopy. If degradation is noticed, a distillation unit will purify the solvent.

37.2.10 Techno-Economic and Lifecycle Analysis

We performed techno-economic analysis for an independent STRAP plant that receives plastic wastes pre-processes it, then applies STRAP for resin extraction entails a few processing lines of separation, starting from ink removal, followed by a few extraction lines, each for a specific resin, as seen in Figure 37.15, extracting the most important plastic resins in the waste blend.

Results are shown in Figure 37.16 for an independent STRAP plant with (i) the annual capital cost, in red, which is the total capital cost multiplied by an annualized factor, and (ii) in blue, the break-even sale price (also production cost) of PE (\$/ton), both vs. plant size (ton/yr) in the range 1,000–16,000 ton/yr. At a plant size of 2,500 ton/yr, the minimum selling price required to return a 10% return is \$750/ton. The real selling price of recycled post-consumer plastics is currently above \$1,500/ton; a STRAP plant of this size is very

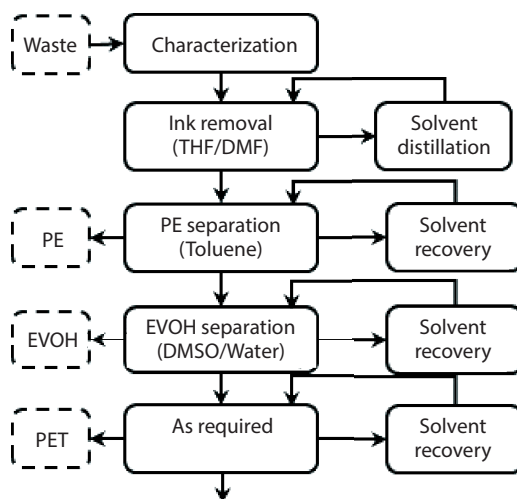


Figure 37.15 Flow diagram for a given waste.

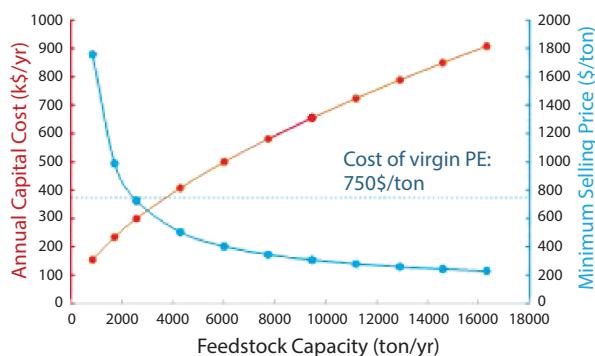


Figure 37.16 The minimum selling price of recycled PE and annualized capital investment of STRAP process for recycling Amcor printed film.

profitable. Profit increases with the size of the plant. A similar analysis for the STRAP-MRF plant with capital and operators' cost are significantly smaller [19].

The cost of the make-up solvent is critical to the STRAP economics. The main source of solvent loss is the absorbed solvent in the targeted resin and the non-dissolved plastics. From lab measurements, we noted that, in average, 1-gr of plastic would trap 1 mL of solvent (or 1-ton traps 1-m³ solvent). Common evaporation-based technologies [20] can reach a solvent recovery rate of 99%. The cost of solvents used is in the range \$4,000–\$6,500/m³. Thus, for 1 ton of plastic produced, one may lose 0.01 m³, which is equivalent to solvent cost \$40–65/ton-plastic. We consider this cost as the maximum possible for solvent recovery in the STRAP process.

We carried out a lifecycle analysis that showed that STRAP (i) has at least 57% [21] lower greenhouse gas emissions than the production of virgin resins, the lowest of any existing recycling plastic technology. The very low carbon-footprint of STRAP would benefit society by recycling plastic wastes, reducing (and eventually replacing) the use of fossil-based

feedstocks, and as shown above, with significant commercial and economic benefits. With an expected continual increase of plastic products and the consequent plastic waste generation, this is a major societal gain.

37.3 Conclusions and Recommendations

The STRAP process extracts specific polymers by preferential dissolution to produce pure resins. This process has been based on molecular dynamic calculations that determine the interactions between the solvents and the polymers and enable accurate predictions and selection of the solvent for a specific polymer. It has been verified experimentally at the bench scale. Lifecycle and techno-economic analyses showed that STRAP (i) has very low specific process energy, and (ii) low capital and operation costs. The following are the main benefits of STRAP: (1) Prevent current disposal in landfills; (2) produce recyclable plastic products; (3) reduce carbon footprint; (4) purchase pure resins at lower costs than those currently purchased. In addition, the societal benefits are (1) reduction of land used for landfills, (2) reduction in GHG generation, (3) reduction in taxation, (4) reduction in ocean plastics, and (5) sustainable (circular) material used. Thus, STRAP would benefit society by recycling mixed plastic wastes and reducing the use of fossil-based feedstocks. With an expected continual increase of plastic products and the consequent plastic waste generation, the continuous STRAP can have major societal gain. The key technological hurdles in scaling up the technology are: (i) integrating the various pilot scale components, (ii) high solvent recovery rate, and (iii) transition to the use of green solvents. We have shown that STRAP can be economically viable at small-medium-large plant sizes. This will allow companies of all sizes that produce plastic components and generate plastic wastes not to anymore landfill their waste and develop recyclable products with low carbon footprint and cost. We foresee three types of users (i) companies who produce plastic products, that generate plastic waste and are currently disposing of it in landfills, (ii) municipalities and companies that are treating post-consumer plastics, and (iii) resin producers that can purchase the STRAP pure resins, at lower costs than currently paid for and distribute them to the market. A sequence of solvents will enable the separation of most of the resins in the waste.

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